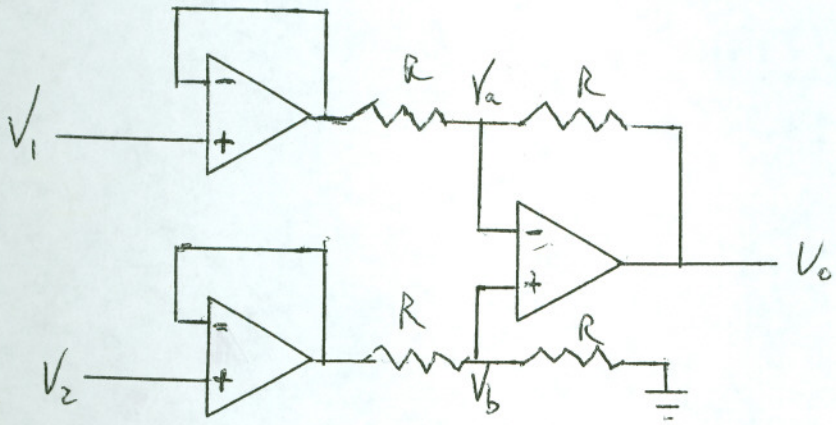
Analytical Cume
September 5, 2003

1. (5 points) Write the three rules for operational amplifiers.

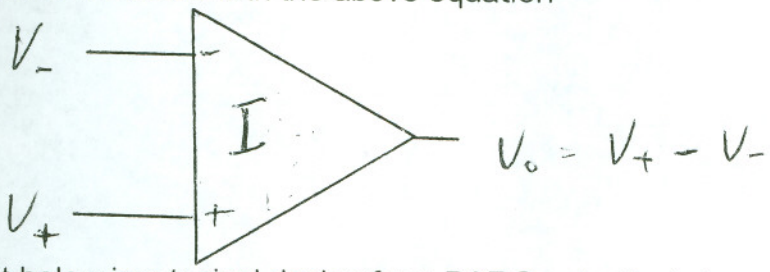
2. (5 points) Write Ohms law.
3. (20 points) Derive the expression:

$$V_o = V_2 - V_1$$

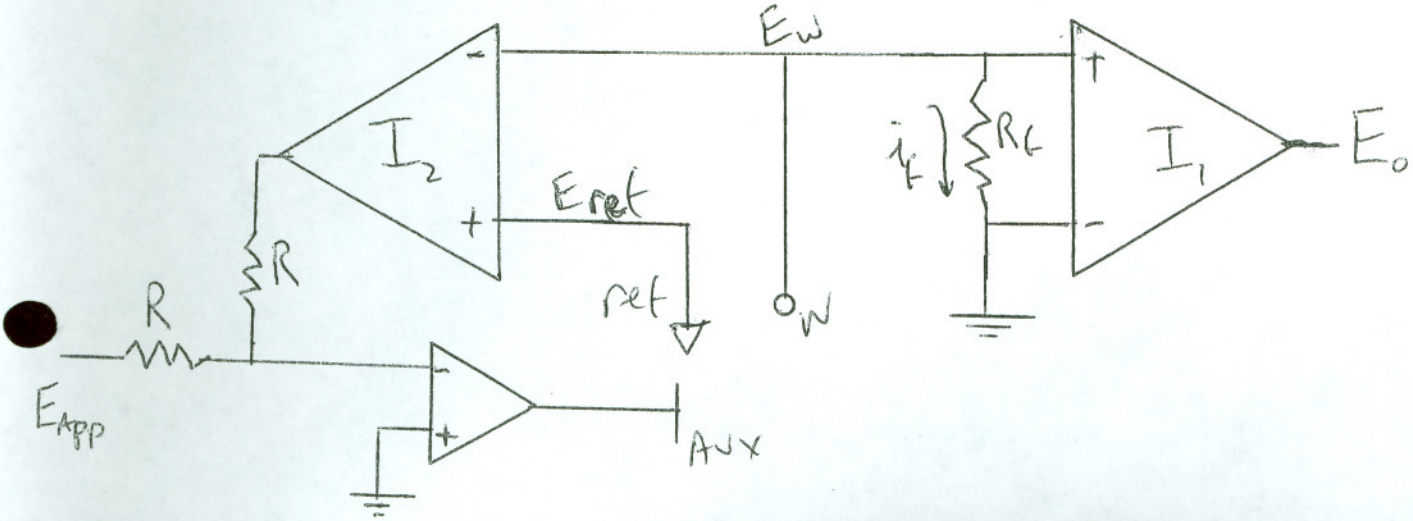
For the following circuit given that all resistance values are equal to 10 kΩ. In your development, use the labels for the potentials shown in the diagram.



Such a circuit is called an instrument amplifier, and is given the symbol:
With the output as indicated with the above equation



4. The circuit below is a typical design for a PARC potentiostat.



The two of the operational amplifiers, marked (I) are instrument amps discussed above. Each question is worth 10 points.

4a. What is E_o in terms of E_w and ground?
4b. What is E_w in terms of R_f and i_f?
4c. What is E_o in terms of R_f and i_f ?
4d. Under what conditions would the potential at the working electrode E_w be zero?
4e. What is the output of the instrument amp (I_2) in terms of E_w and E_{ref}?
4f. How is E_{app} related to the output of instrument amp I_2?
4g. What is the potential difference between the reference and working electrodes?